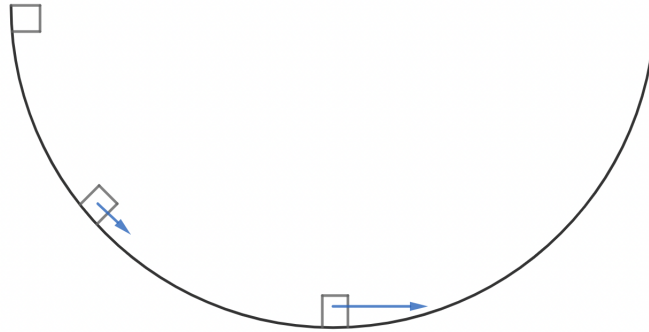


2022A F=ma Exam: Problem 14

Kevin S. Huang



I) By conservation of energy, the speed of the block never decreases since the block is always losing potential energy as it slides down.

II) The horizontal component of the velocity, $v \cos \theta$, never decreases since the speed v is increasing and the angle θ to the horizontal is decreasing ($\cos \theta$ is increasing).

III) The vertical component of the velocity increases and decreases since it's zero at the top and the bottom.

I and II are true so the answer is ☐ C.